Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: DaSilva Dairy Farms LLC #1

Physical address of dairy:

Number and Street	City	County	Zip Code
27398 E Dodds RD	Escalon	San Joaquin	95320

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 04/01/1979

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X207-X210-X001-XXXX

B. OPERATORS

DaSilva, Dennis

Operator name: DaSilva, Dennis

Telephone no.: (209) 000-0000 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
24666 Mariposa RD	Escalon	CA	95320

This operator is responsible for paying permit fees.

C. OWNERS

Da Silva, Farms LLC

Legal owner name: Da Silva, Farms LLC

Telephone no.: (209) 000-0000 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
24666 Mariposa RD	Escalon	CA	95320

DaSilva, Dennis

Legal owner name: DaSilva, Dennis

Telephone no.: (209) 000-0000

Landline	Cellular

Mailing Address Number and Street	City	State	Zip Code
24666 Mariposa RD	Escalon	CA	95320

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	200	0	100	0
Number under roof	1,500	227	0	0	0	0
Maximum number	1,500	228	200	0	100	0
Average number	1,500	227	200	0	0	0
Avg live weight (lbs)	1,300	1,400	1,100	0		

Predominant milk cow breed: Holstein

Average milk production: 77 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 45,144.50 *tons per reporting period*

Total nitrogen from manure: 579,589.48 *lbs per reporting period*

After ammonia losses (30% loss applied): 405,712.64 *lbs per reporting period*

Total phosphorus from manure: 96,809.70 *lbs per reporting period*

Total potassium from manure: 294,162.34 *lbs per reporting period*

Total salt from manure: 758,473.65 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 60,060,000 *gallons*

Total nitrogen generated: 244,419.71 *lbs*

Total phosphorus generated: 27,847.48 *lbs*

Total potassium generated: 277,376.78 *lbs*

Total salt generated: 1,502,450.49 *lbs*

	60,060,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	60,060,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Ag Well/ Bull Ring	Ground water
Ag Well/ F11 & F12	Ground water
Ag Well/ F16 & F17	Ground water
Canal Water	Surface water
Domestic Well/ Dairy Field	Ground water

Source Description	Type
Domestic Well/ Dairy yard	Ground water
Domestic Well/Milk Barn House	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	23	23	0	none	X207-X210-X001-XXXX
Field 10	38	38	0	none	X207-X200-X030-XXXX
Field 11	61	61	2	both	X207-X140-X008-X000
Field 12	80	80	2	both	X207-X140-X008-X000
Field 14	80	80	0	none	X207-X210-X008-XXXX
Field 15	10	10	0	none	X207-X200-X017-XXXX
Field 16	60	60	0	none	X205-X210-X023-XXXX
Field 17	40	40	0	none	X205-X210-X022-XXXX X205-X210-X023-XXXX
Field 2	20	20	0	none	X207-X210-X001-XXXX
Field 3	37	37	2	both	X207-X210-X001-XXXX
Field 4	24	24	2	both	X207-X210-X001-XXXX
Field 5	20	20	2	both	X207-X210-X001-XXXX
Field 7	77	77	2	both	X207-X200-X033-XXXX
Field 8	61	61	2	both	X207-X200-X033-XXXX
Field 9	77	77	2	both	X207-X200-X031-XXXX
Totals for areas that were used for application	437	437	16		
Totals for areas that were not used for application	271	271	0		
Land application area totals	708	708	16		

B. CROPS AND HARVESTS

Field 11

Field name: Field 11

Field 11

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 61 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	1,098.00 *ton*	Dry-weight		82.9	32,100.00	3,030.00	25,300.00		22.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.00	197.61	18.65	155.75	1,378.94

06/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 61 Plant date: 06/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/23/2023	1,830.00 *ton*	Dry-weight		59.5	24,700.00	1,490.00	6,900.00		4.54

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.00	600.21	36.21	167.67	1,103.22

Field 12

Field name: Field 12

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 80 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	1,440.00 *ton*	Dry-weight		79.5	25,000.00	3,090.00	28,200.00		21.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.00	184.50	22.80	208.12	1,557.18

Field 12

06/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 80 Plant date: 06/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/24/2023	2,400.00 *ton*	Dry-weight		62.4	19,400.00	1,260.00	4,200.00		4.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.00	437.66	28.43	94.75	1,105.44

Field 3

Field name: Field 3

10/25/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 37 Plant date: 10/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	666.00 *ton*	Dry-weight		76.8	34,100.00	3,170.00	41,100.00		24.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.00	284.80	26.48	343.27	2,012.83

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	1,110.00 *ton*	Dry-weight		62.8	18,300.00	1,940.00	5,130.00		37.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.00	408.46	43.30	114.50	8,303.04

Field 4

Field name: Field 4

10/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 24 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/07/2023	432.00 *ton*	Dry-weight		78.5	15,200.00	1,930.00	22,300.00		78.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.00	117.65	14.94	172.60	6,075.90

06/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 24 Plant date: 06/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	720.00 *ton*	Dry-weight		64.5	15,600.00	1,370.00	6,870.00		4.96

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.00	332.28	29.18	146.33	1,056.48

Field 5

Field name: Field 5

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/07/2023	360.00 *ton*	Dry-weight		75.4	33,600.00	3,630.00	34,700.00		75.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.00	297.56	32.15	307.30	6,677.42

Field 5

06/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 06/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	600.00 *ton*	Dry-weight		62.2	18,400.00	2,000.00	7,610.00		5.76

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.00	417.31	45.36	172.59	1,306.37

Field 7

Field name: Field 7

10/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 77 Plant date: 10/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	1,386.00 *ton*	Dry-weight		75.5	34,700.00	4,390.00	38,900.00		26.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.00	306.05	38.72	343.10	2,319.66

06/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 77 Plant date: 06/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	2,310.00 *ton*	Dry-weight		65.8	24,300.00	1,850.00	8,970.00		5.78

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.00	498.64	37.96	184.06	1,186.06

Field 8

Field name: Field 8

10/31/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 61 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	1,098.00 *ton*	Dry-weight		77.8	27,200.00	2,290.00	19,400.00		18.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.00	217.38	18.30	155.04	1,478.52

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 61 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/21/2023	1,830.00 *ton*	Dry-weight		67.2	19,200.00	2,050.00	9,850.00		5.78

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.00	377.86	40.34	193.85	1,137.50

Field 9

Field name: Field 9

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 77 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	1,386.00 *ton*	Dry-weight		75.8	33,100.00	1,930.00	17,600.00		16.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.00	288.37	16.81	153.33	1,437.48

Field 9

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 77 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/22/2023	2,310.00 *ton*	Dry-weight		67.5	20,600.00	2,080.00	7,860.00		5.53

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.00	401.70	40.56	153.27	1,078.35

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 11 - 11/03/2022: Oats, silage-soft dough

Field name: Field 11

Crop: Oats, silage-soft dough — Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	42.31	95.97	529.76	3,459.92	488.00 *ton*
Application event totals		42.31	95.97	529.76	3,459.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/08/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	129.28	8.72	86.31	480.18	900,000.00 *gal*
Ag Well/ Bull Ring	Ground water	11.57	0.00	0.00	226.05	2,430,000.00 *gal*
Application event totals		140.85	8.72	86.31	706.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	73.38	6.24	65.01	381.68	900,000.00 *gal*
Ag Well/ Bull Ring	Ground water	11.57	0.00	0.00	226.05	2,430,000.00 *gal*
Application event totals		84.95	6.24	65.01	607.73	

Field 11 - 06/18/2023: Corn, silage

Field name: Field 11

Crop: Corn, silage — Plant date: 06/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 11 - 06/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	26.45	59.98	331.10	2,162.45	305.00 *ton*
Application event totals		26.45	59.98	331.10	2,162.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	117.41	9.99	104.01	610.69	1,440,000.00 *gal*
Canal Water	Surface water	1.51	0.00	0.00	62.33	9,694,753.21 *gal*
Application event totals		118.92	9.99	104.01	673.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	1.44	0.00	0.00	59.22	9,210,015.55 *gal*
Application event totals		1.44	0.00	0.00	59.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	73.87	7.55	81.43	443.24	1,200,000.00 *gal*
Canal Water	Surface water	1.36	0.00	0.00	56.10	8,725,277.89 *gal*
Application event totals		75.23	7.55	81.43	499.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	1.36	0.00	0.00	56.10	8,725,277.89 *gal*
Application event totals		1.36	0.00	0.00	56.10	

Field 11 - 06/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	59.10	6.04	65.14	354.59	960,000.00 *gal*
Canal Water	Surface water	1.30	0.00	0.00	53.50	8,321,329.84 *gal*
Application event totals		60.40	6.04	65.14	408.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	35.20	9.05	79.98	407.13	960,000.00 *gal*
Canal Water	Surface water	1.21	0.00	0.00	49.87	7,755,802.57 *gal*
Application event totals		36.41	9.05	79.98	456.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	26.40	6.79	59.99	305.34	720,000.00 *gal*
Canal Water	Surface water	1.06	0.00	0.00	43.63	6,786,327.25 *gal*
Application event totals		27.46	6.79	59.99	348.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	26.40	6.79	59.99	305.34	720,000.00 *gal*
Canal Water	Surface water	1.06	0.00	0.00	43.63	6,786,327.25 *gal*
Application event totals		27.46	6.79	59.99	348.98	

Field 12 - 11/05/2022: Oats, silage-soft dough

Field name: Field 12

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 12 - 11/05/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/22/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	42.31	95.97	529.76	3,459.92	640.00 *ton*
Application event totals		42.31	95.97	529.76	3,459.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/11/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	131.43	8.86	87.75	488.18	1,200,000.00 *gal*
Ag Well/ Bull Ring	Ground water	9.41	0.00	0.00	183.86	2,592,000.00 *gal*
Application event totals		140.84	8.86	87.75	672.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	55.95	4.76	49.57	291.03	900,000.00 *gal*
Ag Well/ Bull Ring	Ground water	9.41	0.00	0.00	183.86	2,592,000.00 *gal*
Application event totals		65.36	4.76	49.57	474.89	

Field 12 - 06/22/2023: Corn, silage

Field name: Field 12

Crop: Corn, silage

Plant date: 06/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	26.45	59.98	331.10	2,162.45	400.00 *ton*
Application event totals		26.45	59.98	331.10	2,162.45	

Field 12 - 06/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	78.86	8.06	86.92	473.16	1,680,000.00 *gal*
Canal Water	Surface water	1.50	0.00	0.00	61.79	12,603,179.17 *gal*
Application event totals		80.36	8.06	86.92	534.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	1.44	0.00	0.00	59.41	12,118,441.51 *gal*
Application event totals		1.44	0.00	0.00	59.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	67.59	6.91	74.50	405.57	1,440,000.00 *gal*
Canal Water	Surface water	1.15	0.00	0.00	47.53	9,694,753.21 *gal*
Application event totals		68.75	6.91	74.50	453.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	1.15	0.00	0.00	47.53	9,694,753.21 *gal*
Application event totals		1.15	0.00	0.00	47.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	67.59	6.91	74.50	405.57	1,440,000.00 *gal*
Canal Water	Surface water	1.04	0.00	0.00	42.78	8,725,277.89 *gal*
Application event totals		68.63	6.91	74.50	448.34	

Field 12 - 06/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	67.59	6.91	74.50	405.57	1,440,000.00 *gal*
Canal Water	Surface water	1.04	0.00	0.00	42.78	8,725,277.89 *gal*
Application event totals		68.63	6.91	74.50	448.34	

08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	26.84	6.90	60.99	310.43	960,000.00 *gal*
Canal Water	Surface water	0.92	0.00	0.00	38.02	7,755,802.57 *gal*
Application event totals		27.76	6.90	60.99	348.46	

09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	26.84	6.90	60.99	310.43	960,000.00 *gal*
Canal Water	Surface water	0.92	0.00	0.00	38.02	7,755,802.57 *gal*
Application event totals		27.76	6.90	60.99	348.46	

Field 3 - 10/25/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough

Plant date: 10/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	52.89	119.96	662.21	4,324.90	370.00 *ton*
Application event totals		52.89	119.96	662.21	4,324.90	

Field 3 - 10/25/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	142.09	9.58	94.86	527.76	600,000.00 *gal*
Ag Well/ Bull Ring	Ground water	12.72	0.00	0.00	248.46	1,620,000.00 *gal*
Application event totals		154.81	9.58	94.86	776.22	

10/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	36.27	9.32	82.41	419.51	600,000.00 *gal*
Ag Well/ Bull Ring	Ground water	12.72	0.00	0.00	248.46	1,620,000.00 *gal*
Application event totals		48.98	9.32	82.41	667.96	

Field 3 - 05/30/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	26.45	59.98	331.10	2,162.45	185.00 *ton*
Application event totals		26.45	59.98	331.10	2,162.45	

05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	121.79	12.45	134.24	730.75	1,200,000.00 *gal*
Canal Water	Surface water	1.50	0.00	0.00	61.66	5,816,851.92 *gal*
Application event totals		123.29	12.45	134.24	792.41	

Field 3 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	1.25	0.00	0.00	51.38	4,847,376.60 *gal*
Application event totals		1.25	0.00	0.00	51.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	121.79	12.45	134.24	730.75	1,200,000.00 *gal*
Canal Water	Surface water	1.12	0.00	0.00	46.25	4,362,638.94 *gal*
Application event totals		122.91	12.45	134.24	777.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	73.08	7.47	80.55	438.45	720,000.00 *gal*
Canal Water	Surface water	1.12	0.00	0.00	46.25	4,362,638.94 *gal*
Application event totals		74.20	7.47	80.55	484.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	1.00	0.00	0.00	41.11	3,877,901.28 *gal*
Application event totals		1.00	0.00	0.00	41.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	43.52	11.19	98.90	503.41	720,000.00 *gal*
Canal Water	Surface water	1.00	0.00	0.00	41.11	3,877,901.28 *gal*
Application event totals		44.52	11.19	98.90	544.51	

Field 3 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	43.52	11.19	98.90	503.41	720,000.00 *gal*
Canal Water	Surface water	1.00	0.00	0.00	41.11	3,877,901.28 *gal*
Application event totals		44.52	11.19	98.90	544.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.75	0.00	0.00	30.83	2,908,425.96 *gal*
Application event totals		0.75	0.00	0.00	30.83	

Field 4 - 10/26/2022: Oats, silage-soft dough

Field name: Field 4

Crop: Oats, silage-soft dough

Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/27/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well/ Bull Ring	Ground water	15.68	0.00	0.00	306.43	1,296,000.00 *gal*
Application event totals		15.68	0.00	0.00	306.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	124.34	10.58	110.15	646.74	600,000.00 *gal*
Ag Well/ Bull Ring	Ground water	15.68	0.00	0.00	306.43	1,296,000.00 *gal*
Application event totals		140.02	10.58	110.15	953.17	

Field 4 - 06/05/2023: Corn, silage

Field 4 - 06/05/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage

Plant date: 06/05/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	26.45	59.98	331.10	2,162.45	120.00 *ton*
Application event totals		26.45	59.98	331.10	2,162.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	112.66	11.52	124.17	675.95	720,000.00 *gal*
Canal Water	Surface water	1.92	0.00	0.00	79.22	4,847,376.60 *gal*
Application event totals		114.58	11.52	124.17	755.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	1.73	0.00	0.00	71.30	4,362,638.94 *gal*
Application event totals		1.73	0.00	0.00	71.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	75.11	7.68	82.78	450.63	480,000.00 *gal*
Canal Water	Surface water	1.54	0.00	0.00	63.37	3,877,901.28 *gal*
Application event totals		76.64	7.68	82.78	514.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	75.11	7.68	82.78	450.63	480,000.00 *gal*
Canal Water	Surface water	1.54	0.00	0.00	63.37	3,877,901.28 *gal*
Application event totals		76.64	7.68	82.78	514.00	

Field 4 - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	1.54	0.00	0.00	63.37	3,877,901.28 *gal*
Application event totals		1.54	0.00	0.00	63.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	37.55	3.84	41.39	225.32	240,000.00 *gal*
Canal Water	Surface water	1.15	0.00	0.00	47.53	2,908,425.96 *gal*
Application event totals		38.71	3.84	41.39	272.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	37.55	3.84	41.39	225.32	240,000.00 *gal*
Canal Water	Surface water	1.15	0.00	0.00	47.53	2,908,425.96 *gal*
Application event totals		38.71	3.84	41.39	272.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	1.15	0.00	0.00	47.53	2,908,425.96 *gal*
Application event totals		1.15	0.00	0.00	47.53	

Field 5 - 10/28/2022: Oats, silage-soft dough

Field name: Field 5

Crop: Oats, silage-soft dough

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 5 - 10/28/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/29/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	210.29	14.18	140.40	781.09	480,000.00 *gal*
Ag Well/ Bull Ring	Ground water	18.82	0.00	0.00	367.71	1,296,000.00 *gal*
Application event totals		229.11	14.18	140.40	1,148.81	

02/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	119.37	10.15	105.75	620.87	480,000.00 *gal*
Ag Well/ Bull Ring	Ground water	18.82	0.00	0.00	367.71	1,296,000.00 *gal*
Application event totals		138.19	10.15	105.75	988.58	

Field 5 - 06/10/2023: Corn, silage

Field name: Field 5

Crop: Corn, silage

Plant date: 06/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	26.45	59.98	331.10	2,162.45	100.00 *ton*
Application event totals		26.45	59.98	331.10	2,162.45	

05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	90.13	9.21	99.34	540.76	480,000.00 *gal*
Canal Water	Surface water	2.08	0.00	0.00	85.55	4,362,638.94 *gal*
Application event totals		92.20	9.21	99.34	626.31	

Field 5 - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	1.84	0.00	0.00	76.05	3,877,901.28 *gal*
Application event totals		1.84	0.00	0.00	76.05	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	67.59	6.91	74.50	405.57	360,000.00 *gal*
Canal Water	Surface water	1.38	0.00	0.00	57.04	2,908,425.96 *gal*
Application event totals		68.98	6.91	74.50	462.60	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	67.59	6.91	74.50	405.57	360,000.00 *gal*
Canal Water	Surface water	1.38	0.00	0.00	57.04	2,908,425.96 *gal*
Application event totals		68.98	6.91	74.50	462.60	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	26.84	6.90	60.99	310.43	240,000.00 *gal*
Canal Water	Surface water	0.92	0.00	0.00	38.02	1,938,950.64 *gal*
Application event totals		27.76	6.90	60.99	348.46	

08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	26.84	6.90	60.99	310.43	240,000.00 *gal*
Canal Water	Surface water	0.92	0.00	0.00	38.02	1,938,950.64 *gal*
Application event totals		27.76	6.90	60.99	348.46	

Field 5 - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	26.84	6.90	60.99	310.43	240,000.00 *gal*
Canal Water	Surface water	0.92	0.00	0.00	38.02	1,938,950.64 *gal*
Application event totals		27.76	6.90	60.99	348.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	0.92	0.00	0.00	38.02	1,938,950.64 *gal*
Application event totals		0.92	0.00	0.00	38.02	

Field 7 - 10/30/2022: Oats, silage-soft dough

Field name: Field 7

Crop: Oats, silage-soft dough Plant date: 10/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	52.89	119.96	662.21	4,324.90	770.00 *ton*
Application event totals		52.89	119.96	662.21	4,324.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	136.55	9.21	91.17	507.20	1,200,000.00 *gal*
Ag Well/ Bull Ring	Ground water	2.44	0.00	0.00	47.76	648,000.00 *gal*
Application event totals		139.00	9.21	91.17	554.96	

Field 7 - 10/30/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	77.51	6.59	68.67	403.16	1,200,000.00 *gal*
Ag Well/ Bull Ring	Ground water	2.44	0.00	0.00	47.76	648,000.00 *gal*
Application event totals		79.95	6.59	68.67	450.92	

Field 7 - 06/13/2023: Corn, silage

Field name: Field 7

Crop: Corn, silage

Plant date: 06/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	26.45	59.98	331.10	2,162.45	385.00 *ton*
Application event totals		26.45	59.98	331.10	2,162.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	93.01	7.91	82.40	483.79	1,440,000.00 *gal*
Canal Water	Surface water	1.56	0.00	0.00	64.20	12,603,179.17 *gal*
Application event totals		94.57	7.91	82.40	547.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	1.50	0.00	0.00	61.73	12,118,441.51 *gal*
Application event totals		1.50	0.00	0.00	61.73	

Field 7 - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	70.23	7.18	77.41	421.37	1,440,000.00 *gal*
Canal Water	Surface water	1.44	0.00	0.00	59.26	11,633,703.85 *gal*
Application event totals		71.67	7.18	77.41	480.63	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	1.44	0.00	0.00	59.26	11,633,703.85 *gal*
Application event totals		1.44	0.00	0.00	59.26	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	70.23	7.18	77.41	421.37	1,440,000.00 *gal*
Canal Water	Surface water	1.20	0.00	0.00	49.38	9,694,753.21 *gal*
Application event totals		71.43	7.18	77.41	470.75	

08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	70.23	7.18	77.41	421.37	1,440,000.00 *gal*
Canal Water	Surface water	1.20	0.00	0.00	49.38	9,694,753.21 *gal*
Application event totals		71.43	7.18	77.41	470.75	

08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	46.82	4.79	51.60	280.91	960,000.00 *gal*
Canal Water	Surface water	0.96	0.00	0.00	39.51	7,755,802.57 *gal*
Application event totals		47.78	4.79	51.60	320.42	

Field 7 - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	27.88	7.17	63.36	322.53	960,000.00 *gal*
Canal Water	Surface water	0.96	0.00	0.00	39.51	7,755,802.57 *gal*
Application event totals		28.84	7.17	63.36	362.03	

Field 8 - 10/31/2022: Oats, silage-soft dough

Field name: Field 8

Crop: Oats, silage-soft dough

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	52.89	119.96	662.21	4,324.90	610.00 *ton*
Application event totals		52.89	119.96	662.21	4,324.90	

11/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	129.28	8.72	86.31	480.18	900,000.00 *gal*
Ag Well/ Bull Ring	Ground water	3.86	0.00	0.00	75.35	810,000.00 *gal*
Application event totals		133.14	8.72	86.31	555.53	

02/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	73.38	6.24	65.01	381.68	900,000.00 *gal*
Ag Well/ Bull Ring	Ground water	3.86	0.00	0.00	75.35	810,000.00 *gal*
Application event totals		77.24	6.24	65.01	457.03	

Field 8 - 06/15/2023: Corn, silage

Field 8 - 06/15/2023: Corn, silage

Field name: Field 8

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	26.45	59.98	331.10	2,162.45	305.00 *ton*
Application event totals		26.45	59.98	331.10	2,162.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	88.65	9.06	97.71	531.89	1,440,000.00 *gal*
Canal Water	Surface water	1.36	0.00	0.00	56.10	8,725,277.89 *gal*
Application event totals		90.01	9.06	97.71	587.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	1.29	0.00	0.00	52.98	8,240,540.23 *gal*
Application event totals		1.29	0.00	0.00	52.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	73.87	7.55	81.43	443.24	1,200,000.00 *gal*
Canal Water	Surface water	1.21	0.00	0.00	49.87	7,755,802.57 *gal*
Application event totals		75.08	7.55	81.43	493.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	73.87	7.55	81.43	443.24	1,200,000.00 *gal*
Canal Water	Surface water	1.21	0.00	0.00	49.87	7,755,802.57 *gal*
Application event totals		75.08	7.55	81.43	493.11	

Field 8 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	35.20	9.05	79.98	407.13	960,000.00 *gal*
Canal Water	Surface water	1.21	0.00	0.00	49.87	7,755,802.57 *gal*
Application event totals		36.41	9.05	79.98	456.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	35.20	9.05	79.98	407.13	960,000.00 *gal*
Canal Water	Surface water	1.21	0.00	0.00	49.87	7,755,802.57 *gal*
Application event totals		36.41	9.05	79.98	456.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	26.40	6.79	59.99	305.34	720,000.00 *gal*
Canal Water	Surface water	1.06	0.00	0.00	43.63	6,786,327.25 *gal*
Application event totals		27.46	6.79	59.99	348.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	26.40	6.79	59.99	305.34	720,000.00 *gal*
Canal Water	Surface water	1.06	0.00	0.00	43.63	6,786,327.25 *gal*
Application event totals		27.46	6.79	59.99	348.98	

Field 9 - 11/01/2022: Oats, silage-soft dough

Field name: Field 9

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 9 - 11/01/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	52.89	119.96	662.21	4,324.90	770.00 *ton*
Application event totals		52.89	119.96	662.21	4,324.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/08/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	136.55	9.21	91.17	507.20	1,200,000.00 *gal*
Ag Well/ Bull Ring	Ground water	9.16	0.00	0.00	179.08	2,430,000.00 *gal*
Application event totals		145.72	9.21	91.17	686.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	77.51	6.59	68.67	403.16	1,200,000.00 *gal*
Ag Well/ Bull Ring	Ground water	9.16	0.00	0.00	179.08	2,430,000.00 *gal*
Application event totals		86.68	6.59	68.67	582.24	

Field 9 - 06/14/2023: Corn, silage

Field name: Field 9

Crop: Corn, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure	Corral solids	26.45	59.98	331.10	2,162.45	385.00 *ton*
Application event totals		26.45	59.98	331.10	2,162.45	

Field 9 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	81.93	8.38	90.31	491.60	1,680,000.00 *gal*
Canal Water	Surface water	1.56	0.00	0.00	64.20	12,603,179.17 *gal*
Application event totals		83.49	8.38	90.31	555.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Water	Surface water	1.50	0.00	0.00	61.73	12,118,441.51 *gal*
Application event totals		1.50	0.00	0.00	61.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	70.23	7.18	77.41	421.37	1,440,000.00 *gal*
Canal Water	Surface water	1.44	0.00	0.00	59.26	11,633,703.85 *gal*
Application event totals		71.67	7.18	77.41	480.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	70.23	7.18	77.41	421.37	1,440,000.00 *gal*
Canal Water	Surface water	1.44	0.00	0.00	59.26	11,633,703.85 *gal*
Application event totals		71.67	7.18	77.41	480.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	41.82	10.75	95.04	483.79	1,440,000.00 *gal*
Canal Water	Surface water	1.20	0.00	0.00	49.38	9,694,753.21 *gal*
Application event totals		43.02	10.75	95.04	533.18	

Field 9 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	41.82	10.75	95.04	483.79	1,440,000.00 *gal*
Canal Water	Surface water	1.20	0.00	0.00	49.38	9,694,753.21 *gal*
Application event totals		43.02	10.75	95.04	533.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	27.88	7.17	63.36	322.53	960,000.00 *gal*
Canal Water	Surface water	0.96	0.00	0.00	39.51	7,755,802.57 *gal*
Application event totals		28.84	7.17	63.36	362.03	

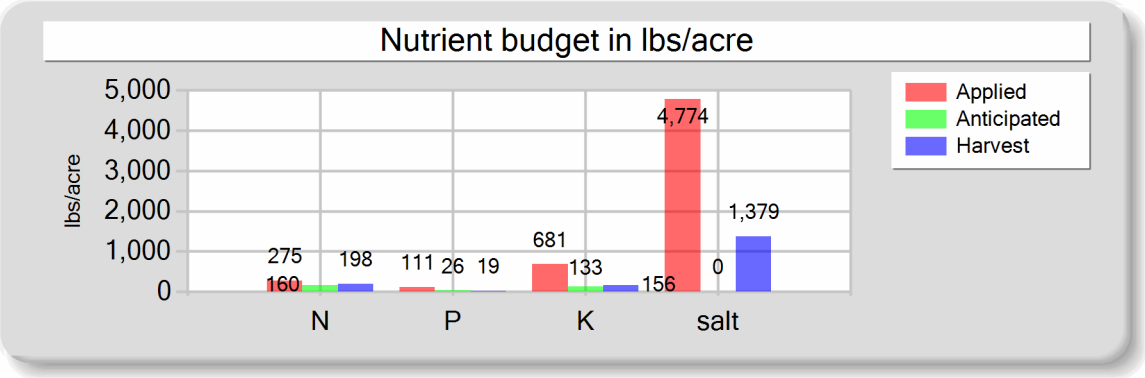
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	27.88	7.17	63.36	322.53	960,000.00 *gal*
Canal Water	Surface water	0.96	0.00	0.00	39.51	7,755,802.57 *gal*
Application event totals		28.84	7.17	63.36	362.03	

B. NUTRIENT BUDGET

Field 11 - 11/03/2022: Oats, silage-soft dough

Field name: Field 11 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	42.31	95.97	529.76	3,459.92
Process wastewater	202.66	14.96	151.32	861.86
Fresh water	23.14	0.00	0.00	452.11
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	275.11	110.93	681.08	4,773.89
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	197.61	18.65	155.75	1,378.94
Nutrient balance	77.50	92.28	525.34	3,394.94
Applied to removed ratio	1.39	5.95	4.37	3.46

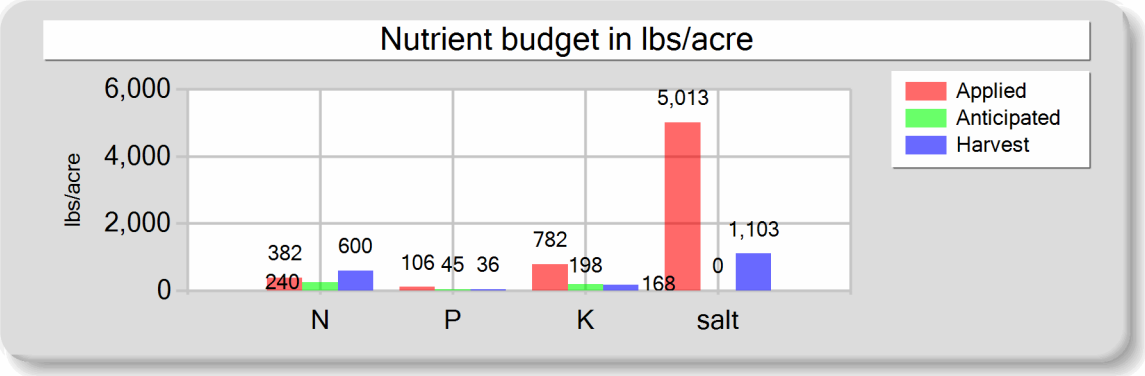
Fresh water applied	
4,860,000.00	*gallons*
178.98	*acre-inches*
2.93	*inches/acre*

Process wastewater applied	
1,800,000.00	*gallons*
66.29	*acre-inches*
1.09	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 11 - 06/18/2023: Corn, silage

Field name: Field 11 Crop: Corn, silage Plant date: 06/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	26.45	59.98	331.10	2,162.45
Process wastewater	338.37	46.20	450.53	2,426.34
Fresh water	10.29	0.00	0.00	424.40
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	382.11	106.18	781.63	5,013.19
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	600.21	36.21	167.67	1,103.22
Nutrient balance	-218.10	69.98	613.96	3,909.97
Applied to removed ratio	0.64	2.93	4.66	4.54

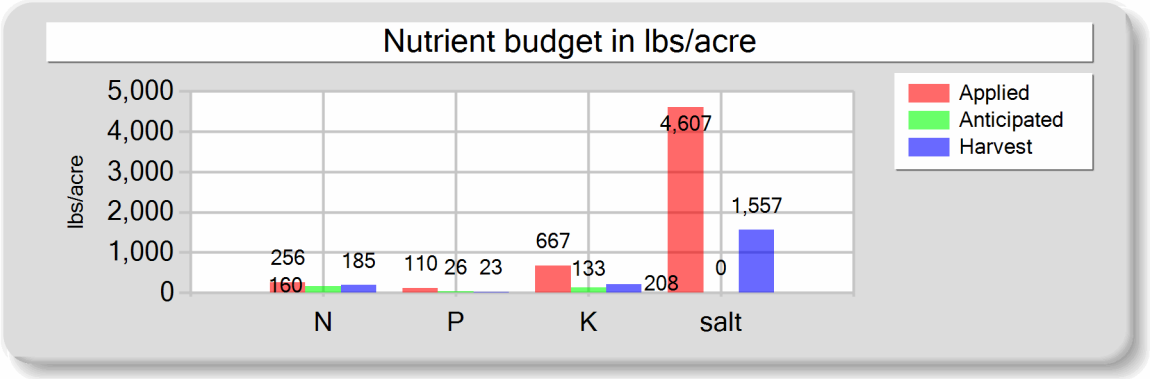
Fresh water applied	
66,005,111.45	*gallons*
2,430.74	*acre-inches*
39.85	*inches/acre*

Process wastewater applied	
6,000,000.00	*gallons*
220.96	*acre-inches*
3.62	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 12 - 11/05/2022: Oats, silage-soft dough

Field name: Field 12 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	42.31	95.97	529.76	3,459.92
Process wastewater	187.39	13.62	137.32	779.21
Fresh water	18.82	0.00	0.00	367.71
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	255.52	109.59	667.08	4,606.85
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	184.50	22.80	208.12	1,557.18
Nutrient balance	71.02	86.79	458.97	3,049.67
Applied to removed ratio	1.38	4.81	3.21	2.96

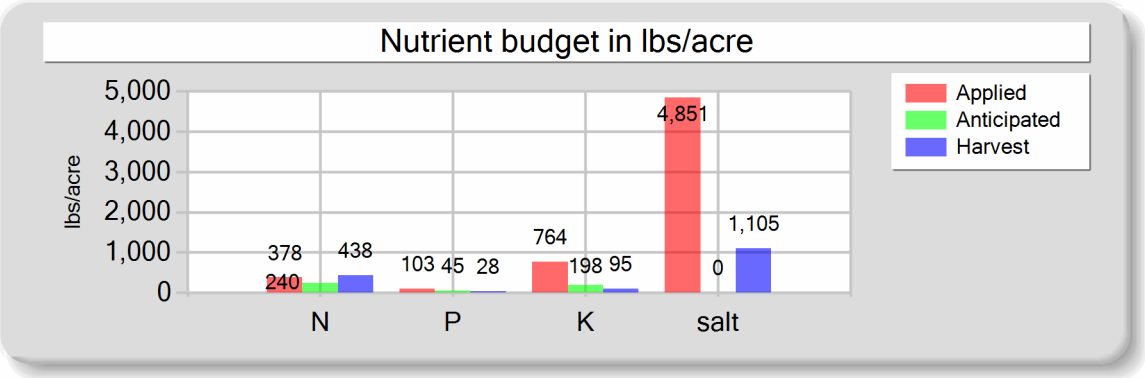
Fresh water applied	
5,184,000.00	*gallons*
190.91	*acre-inches*
2.39	*inches/acre*

Process wastewater applied	
2,100,000.00	*gallons*
77.34	*acre-inches*
0.97	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 12 - 06/22/2023: Corn, silage

Field name: Field 12 Crop: Corn, silage Plant date: 06/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	26.45	59.98	331.10	2,162.45
Process wastewater	335.32	42.59	432.40	2,310.73
Fresh water	9.17	0.00	0.00	377.87
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	377.93	102.57	763.51	4,851.05
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	437.66	28.43	94.75	1,105.44
Nutrient balance	-59.73	74.15	668.76	3,745.61
Applied to removed ratio	0.86	3.61	8.06	4.39

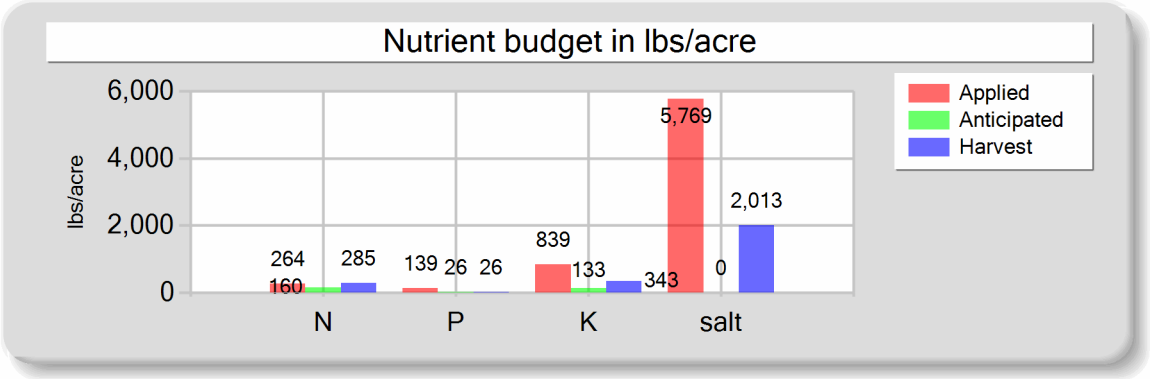
Fresh water applied	
77,073,288.02	*gallons*
2,838.35	*acre-inches*
35.48	*inches/acre*

Process wastewater applied	
7,920,000.00	*gallons*
291.67	*acre-inches*
3.65	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 10/25/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 10/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	52.89	119.96	662.21	4,324.90
Process wastewater	178.36	18.90	177.27	947.27
Fresh water	25.43	0.00	0.00	496.91
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	263.68	138.87	839.48	5,769.08
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	284.80	26.48	343.27	2,012.83
Nutrient balance	-21.12	112.39	496.21	3,756.25
Applied to removed ratio	0.93	5.25	2.45	2.87

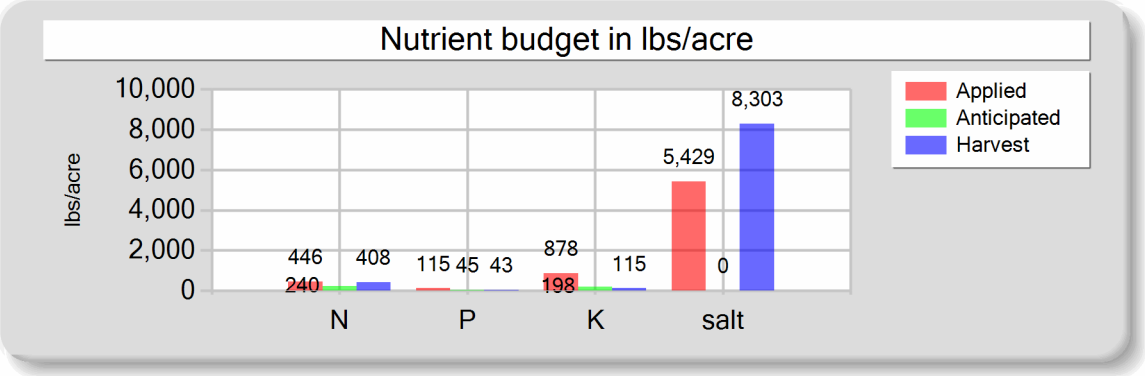
Fresh water applied	
3,240,000.00	*gallons*
119.32	*acre-inches*
3.22	*inches/acre*

Process wastewater applied	
1,200,000.00	*gallons*
44.19	*acre-inches*
1.19	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 05/30/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	26.45	59.98	331.10	2,162.45
Process wastewater	403.70	54.75	546.82	2,906.77
Fresh water	8.72	0.00	0.00	359.69
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	445.87	114.73	877.92	5,428.91
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	408.46	43.30	114.50	8,303.04
Nutrient balance	37.41	71.43	763.42	-2,874.13
Applied to removed ratio	1.09	2.65	7.67	0.65

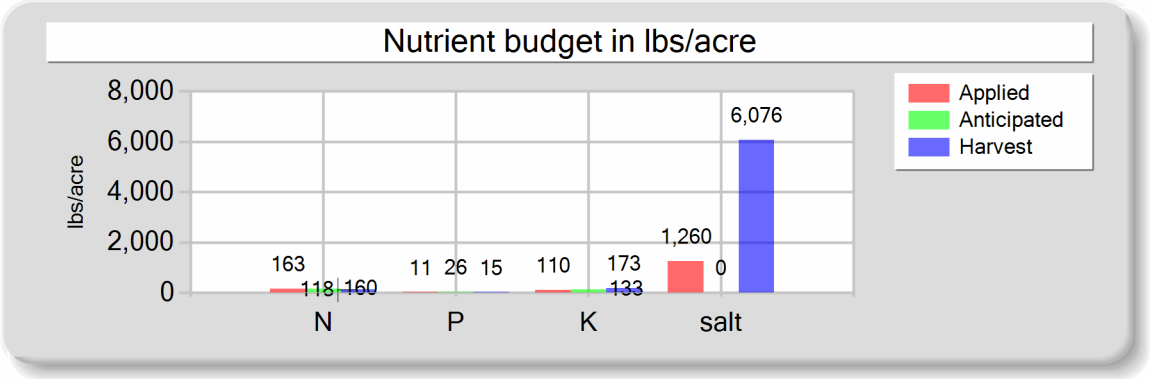
Fresh water applied	
33,931,636.20	*gallons*
1,249.59	*acre-inches*
33.77	*inches/acre*

Process wastewater applied	
4,560,000.00	*gallons*
167.93	*acre-inches*
4.54	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 10/26/2022: Oats, silage-soft dough

Field name: Field 4 Crop: Oats, silage-soft dough Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	124.34	10.58	110.15	646.74
Fresh water	31.36	0.00	0.00	612.86
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	162.70	10.58	110.15	1,259.59
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	117.65	14.94	172.60	6,075.90
Nutrient balance	45.06	-4.36	-62.45	-4,816.31
Applied to removed ratio	1.38	0.71	0.64	0.21

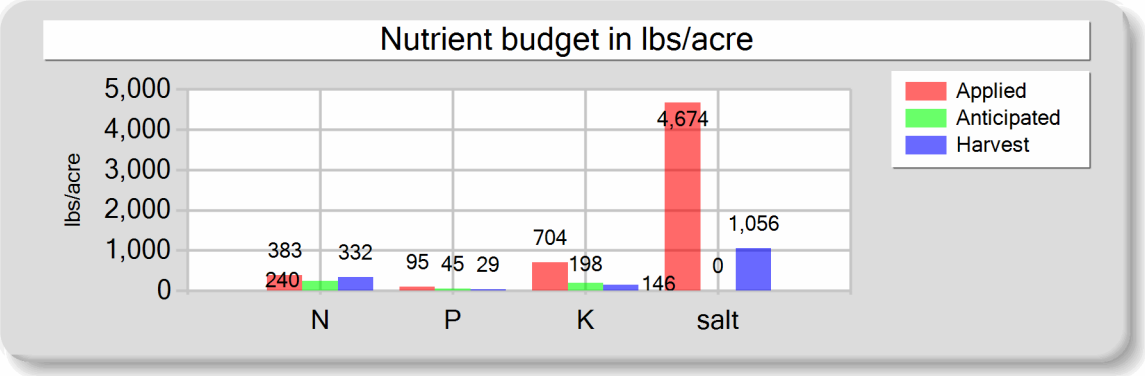
Fresh water applied	
2,592,000.00	*gallons*
95.45	*acre-inches*
3.98	*inches/acre*

Process wastewater applied	
600,000.00	*gallons*
22.10	*acre-inches*
0.92	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 06/05/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 06/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	26.45	59.98	331.10	2,162.45
Process wastewater	337.97	34.55	372.52	2,027.84
Fresh water	11.72	0.00	0.00	483.23
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	383.14	94.53	703.62	4,673.51
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	332.28	29.18	146.33	1,056.48
Nutrient balance	50.86	65.35	557.29	3,617.03
Applied to removed ratio	1.15	3.24	4.81	4.42

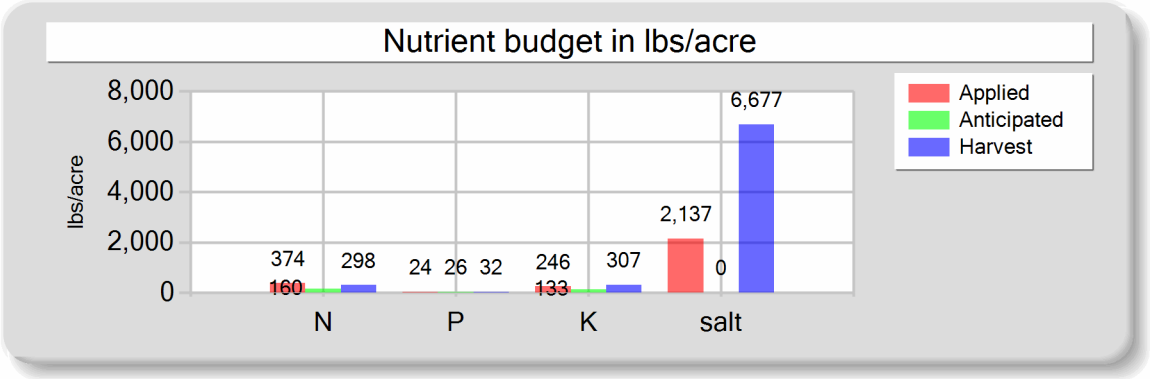
Fresh water applied	
29,568,997.26	*gallons*
1,088.93	*acre-inches*
45.37	*inches/acre*

Process wastewater applied	
2,160,000.00	*gallons*
79.55	*acre-inches*
3.31	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 10/28/2022: Oats, silage-soft dough

Field name: Field 5 Crop: Oats, silage-soft dough Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	329.66	24.33	246.14	1,401.96
Fresh water	37.64	0.00	0.00	735.43
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	374.30	24.33	246.14	2,137.39
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	297.56	32.15	307.30	6,677.42
Nutrient balance	76.74	-7.81	-61.16	-4,540.04
Applied to removed ratio	1.26	0.76	0.80	0.32

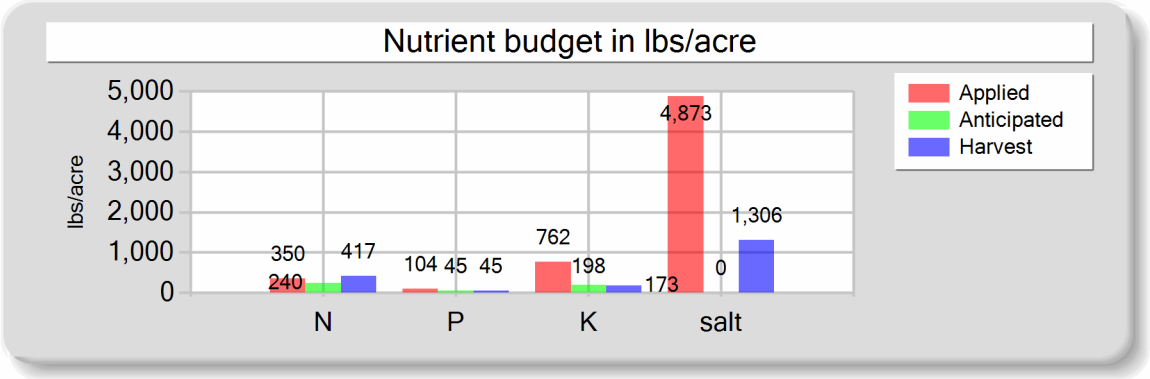
Fresh water applied	
2,592,000.00	*gallons*
95.45	*acre-inches*
4.77	*inches/acre*

Process wastewater applied	
960,000.00	*gallons*
35.35	*acre-inches*
1.77	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 06/10/2023: Corn, silage

Field name: Field 5 Crop: Corn, silage Plant date: 06/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	26.45	59.98	331.10	2,162.45
Process wastewater	305.83	43.73	431.30	2,283.19
Fresh water	10.38	0.00	0.00	427.77
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	349.65	103.71	762.41	4,873.42
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	417.31	45.36	172.59	1,306.37
Nutrient balance	-67.66	58.35	589.81	3,567.05
Applied to removed ratio	0.84	2.29	4.42	3.73

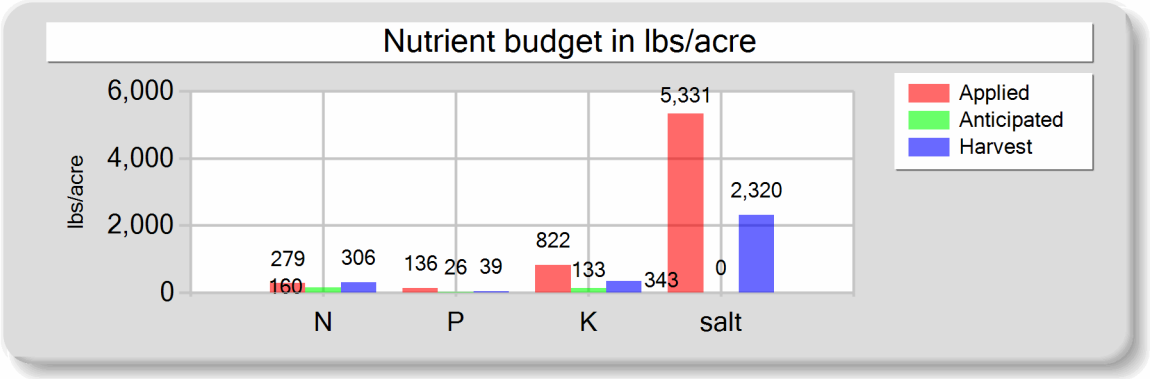
Fresh water applied	
21,813,194.70	*gallons*
803.31	*acre-inches*
40.17	*inches/acre*

Process wastewater applied	
1,920,000.00	*gallons*
70.71	*acre-inches*
3.54	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 7 - 10/30/2022: Oats, silage-soft dough

Field name: Field 7 Crop: Oats, silage-soft dough Plant date: 10/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	52.89	119.96	662.21	4,324.90
Process wastewater	214.07	15.80	159.83	910.36
Fresh water	4.89	0.00	0.00	95.51
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	278.84	135.76	822.04	5,330.77
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	306.05	38.72	343.10	2,319.66
Nutrient balance	-27.21	97.04	478.94	3,011.11
Applied to removed ratio	0.91	3.51	2.40	2.30

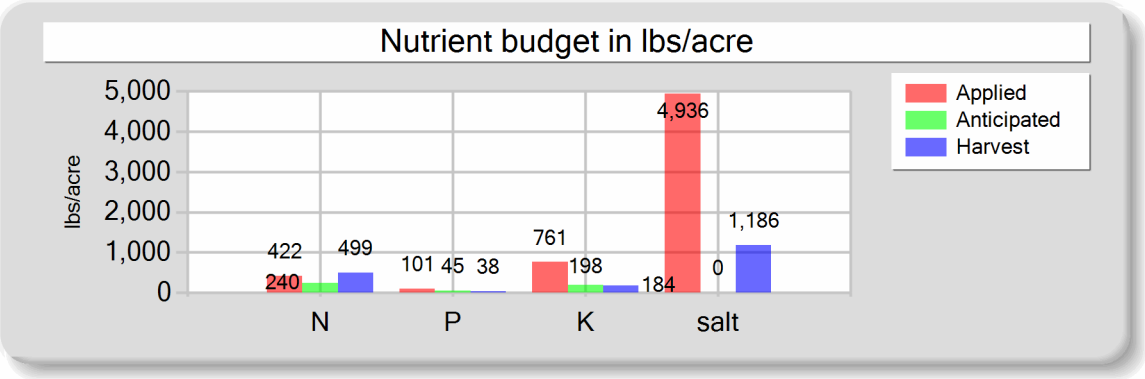
Fresh water applied	
1,296,000.00	*gallons*
47.73	*acre-inches*
0.62	*inches/acre*

Process wastewater applied	
2,400,000.00	*gallons*
88.38	*acre-inches*
1.15	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 7 - 06/13/2023: Corn, silage

Field name: Field 7 Crop: Corn, silage Plant date: 06/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	26.45	59.98	331.10	2,162.45
Process wastewater	378.40	41.40	429.59	2,351.34
Fresh water	10.24	0.00	0.00	422.22
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	422.09	101.38	760.69	4,936.01
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	498.64	37.96	184.06	1,186.06
Nutrient balance	-76.55	63.42	576.63	3,749.95
Applied to removed ratio	0.85	2.67	4.13	4.16

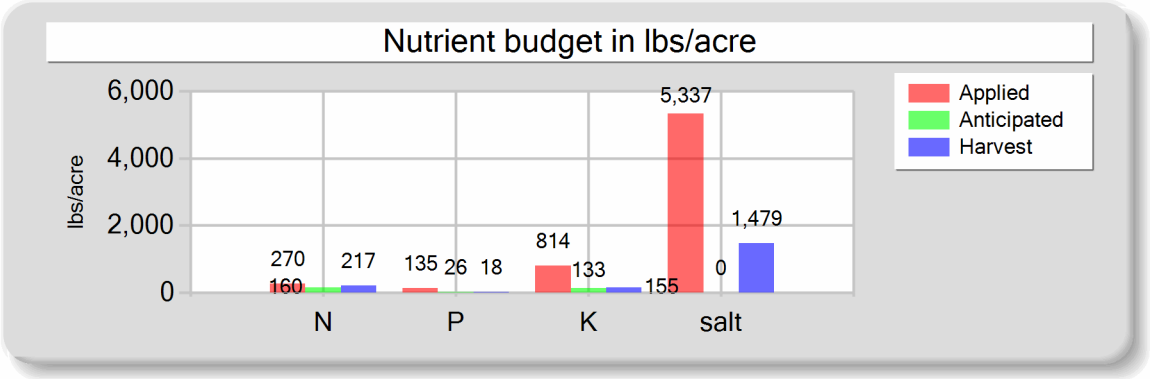
Fresh water applied	
82,890,139.94	*gallons*
3,052.56	*acre-inches*
39.64	*inches/acre*

Process wastewater applied	
7,680,000.00	*gallons*
282.83	*acre-inches*
3.67	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 8 - 10/31/2022: Oats, silage-soft dough

Field name: Field 8 Crop: Oats, silage-soft dough Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	52.89	119.96	662.21	4,324.90
Process wastewater	202.66	14.96	151.32	861.86
Fresh water	7.71	0.00	0.00	150.70
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	270.26	134.92	813.52	5,337.46
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	217.38	18.30	155.04	1,478.52
Nutrient balance	52.88	116.62	658.48	3,858.94
Applied to removed ratio	1.24	7.37	5.25	3.61

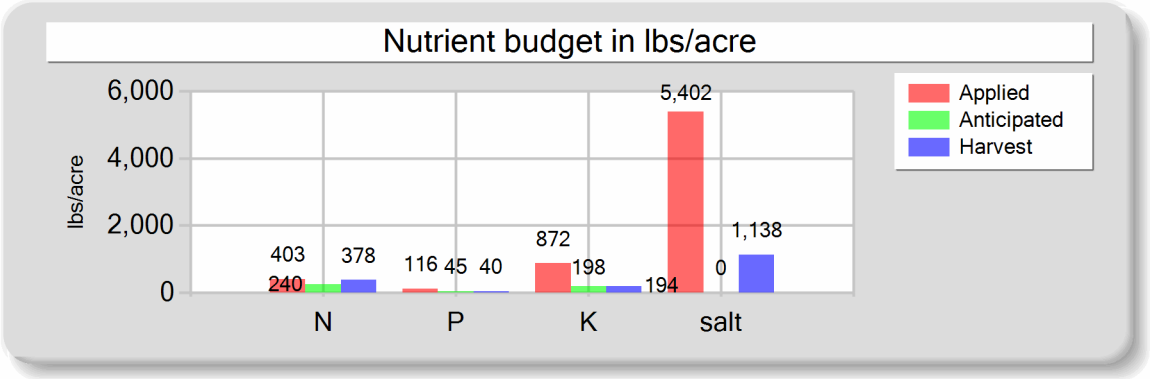
Fresh water applied	
1,620,000.00	*gallons*
59.66	*acre-inches*
0.98	*inches/acre*

Process wastewater applied	
1,800,000.00	*gallons*
66.29	*acre-inches*
1.09	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 8 - 06/15/2023: Corn, silage

Field name: Field 8 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	26.45	59.98	331.10	2,162.45
Process wastewater	359.58	55.84	540.49	2,843.32
Fresh water	9.60	0.00	0.00	395.83
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	402.63	115.82	871.60	5,401.60
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	377.86	40.34	193.85	1,137.50
Nutrient balance	24.78	75.47	677.75	4,264.09
Applied to removed ratio	1.07	2.87	4.50	4.75

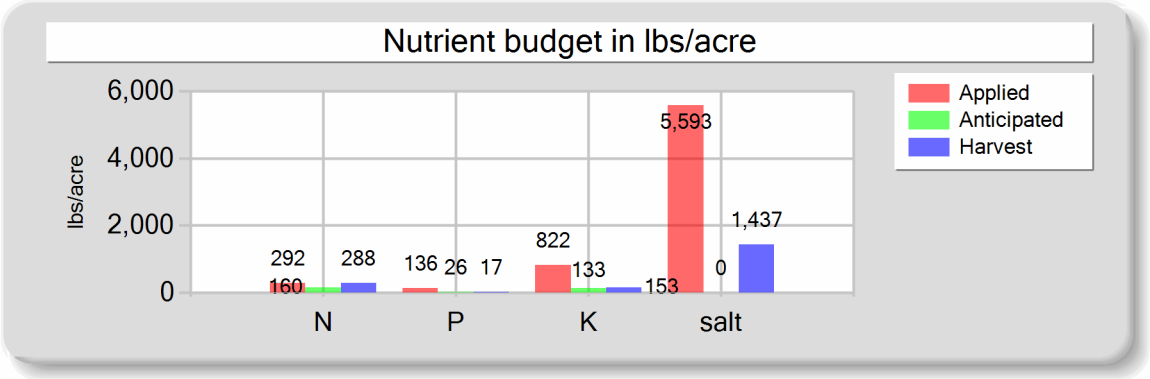
Fresh water applied	
61,561,682.90	*gallons*
2,267.11	*acre-inches*
37.17	*inches/acre*

Process wastewater applied	
7,200,000.00	*gallons*
265.15	*acre-inches*
4.35	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 9 - 11/01/2022: Oats, silage-soft dough

Field name: Field 9 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	52.89	119.96	662.21	4,324.90
Process wastewater	214.07	15.80	159.83	910.36
Fresh water	18.33	0.00	0.00	358.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	292.29	135.76	822.04	5,593.43
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	288.37	16.81	153.33	1,437.48
Nutrient balance	3.92	118.95	668.71	4,155.95
Applied to removed ratio	1.01	8.07	5.36	3.89

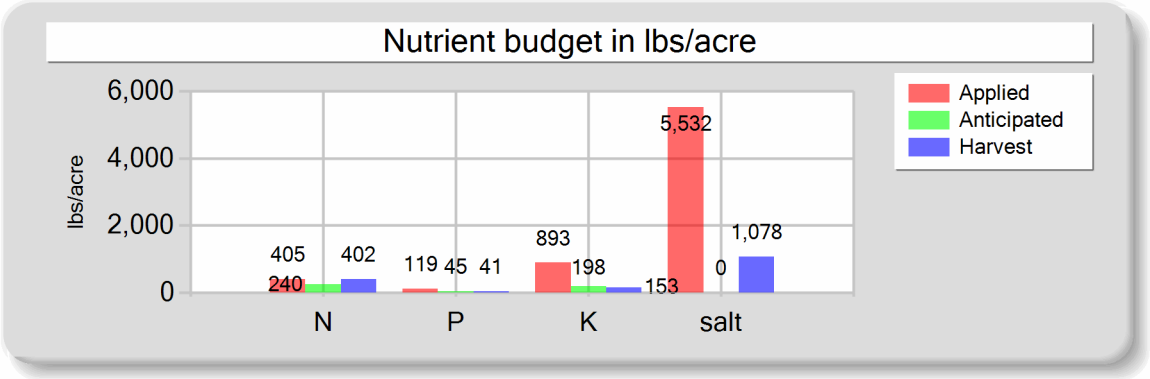
Fresh water applied	
4,860,000.00	*gallons*
178.98	*acre-inches*
2.32	*inches/acre*

Process wastewater applied	
2,400,000.00	*gallons*
88.38	*acre-inches*
1.15	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 9 - 06/14/2023: Corn, silage

Field name: Field 9 Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	26.45	59.98	331.10	2,162.45
Process wastewater	361.80	58.58	561.93	2,946.98
Fresh water	10.24	0.00	0.00	422.22
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	405.49	118.56	893.03	5,531.64
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	401.70	40.56	153.27	1,078.35
Nutrient balance	3.79	78.00	739.76	4,453.29
Applied to removed ratio	1.01	2.92	5.83	5.13

Fresh water applied	
82,890,139.94	*gallons*
3,052.56	*acre-inches*
39.64	*inches/acre*

Process wastewater applied	
9,360,000.00	*gallons*
344.70	*acre-inches*
4.48	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Manure 2102513-01

Sample and source description: Manure 2102513-01

Sample date: 10/22/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 29.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,730.00	8,460.00	46,700.00	21,000.00	10,000.00	12,000.00	4,880.00	6,160.00		30.50
DL	0.50	0.50	0.50	5.00	5.00	5.00	1.00	0.50		0.10

B. PROCESS WASTEWATER ANALYSES

Lagoon Water 2302042-01

Sample and source description: Lagoon Water 2302042-01

Sample date: 01/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,050.00	1,050.00			70.80	701.00								7,000.00	3,900
DL	1.20	0.60			0.10	10.00								1.00	1

Lagoon Water 2303404-01

Sample and source description: Lagoon Water 2303404-01

Sample date: 03/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	596.00	596.00			50.70	528.00								5,230.00	3,100
DL	0.60	10.00			0.10	10.00								1.00	1

Lagoon Water 2307247-01

Sample and source description: Lagoon Water 2307247-01

Sample date: 07/17/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	450.00	448.00			46.00	496.00								5,030.00	2,700
DL	0.60	0.60			0.10	10.00								1.00	1

Lagoon Water 2310050-01

Sample and source description: Lagoon Water 2310050-01

Sample date: 09/29/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	268.00	268.00			68.90	609.00								5,460.00	3,100
DL	0.60	0.60			0.10	10.00								1.00	1

C. FRESH WATER ANALYSES

Ag Well/ Bull Ring

2401174-06

Sample description: 2401174-06

Sample date: 01/12/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	34.80		34.60								1,140.00	680
DL	1.40		0.40								1.00	1

Ag Well/ F11 & F12

Ag Well/ F11 & F12

2401174-04

Sample description: 2401174-04

Sample date: 01/12/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	35.50		35.50								1,130.00	680
DL	1.40		0.40								1.00	1

Ag Well/ F16 & F17

2401174-05

Sample description: 2401174-05

Sample date: 01/12/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	35.50		34.90								1,170.00	730
DL	1.40		0.40								1.00	1

Canal Water

2310048-01

Sample description: 2310048-01

Sample date: 09/29/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.14										60.20	47
DL	0.40										1.00	1

Domestic Well/ Dairy Field

Domestic Well/ Dairy Field

2401174-01

Sample description: 2401174-01

Sample date: 01/12/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	36.10										1,100.00	
DL	0.40										1.00	

Domestic Well/ Dairy yard

2401174-03

Sample description: 2401174-03

Sample date: 01/12/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	37.30										1,180.00	
DL	0.40										1.00	

Domestic Well/Milk Barn House

2401174-02

Sample description: 2401174-02

Sample date: 01/12/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	35.70										1,170.00	
DL	0.40										1.00	

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field 11 - 11/03/2022: Oats, silage-soft dough

2305231-03

Sample and source description: 2305231-03

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 82.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	32,100.00	3,030.00	25,300.00		22.40
DL	0.50	0.50	0.50		0.10

Field 11 - 06/18/2023: Corn, silage

2310432-01

Sample and source description: 2310432-01

Sample date: 10/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,100.00	1,770.00	8,080.00		5.89
DL	0.50	0.50	0.50		0.10

2310432-02

Sample and source description: 2310432-02

Sample date: 10/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,700.00	1,490.00	6,900.00		4.54
DL	0.50	0.50	0.50		0.10

Field 12 - 11/05/2022: Oats, silage-soft dough

2305231-04

Sample and source description: 2305231-04

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,700.00	2,600.00	25,600.00		21.70
DL	0.50	0.50	0.50		0.10

2305231-05

Sample and source description: 2305231-05

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 79.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,000.00	3,090.00	28,200.00		21.10
DL	0.50	0.50	0.50		0.10

Field 12 - 06/22/2023: Corn, silage

2310432-03

Sample and source description: 2310432-03

Sample date: 10/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 27.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,300.00	1,010.00	4,010.00		78.50
DL	0.50	0.50	0.50		0.10

Field 12 - 06/22/2023: Corn, silage

2310432-04)

Sample and source description: 2310432-04)

Sample date: 10/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,400.00	1,260.00	4,200.00		4.90
DL	0.50	0.50	0.50		0.10

Field 3 - 10/25/2022: Oats, silage-soft dough

2305230-01

Sample and source description: 2305230-01

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	34,100.00	3,170.00	41,100.00		24.10
DL	2.50	0.50	0.50		0.10

Field 3 - 05/30/2023: Corn, silage

2310431-01

Sample and source description: 2310431-01

Sample date: 10/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,300.00	1,940.00	5,130.00		37.20
DL	0.50	0.50	0.50		0.10

Field 4 - 10/26/2022: Oats, silage-soft dough

2305230-02

Sample and source description: 2305230-02

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,200.00	1,930.00	22,300.00		78.50
DL	0.50	0.50	0.50		0.10

Field 4 - 06/05/2023: Corn, silage

2310431-02

Sample and source description: 2310431-02

Sample date: 10/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,600.00	1,370.00	6,870.00		4.96
DL	0.50	0.50	0.50		0.10

Field 5 - 10/28/2022: Oats, silage-soft dough

2305230-03

Sample and source description: 2305230-03

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	33,600.00	3,630.00	34,700.00		75.40
DL	0.50	0.50	0.50		0.10

Field 5 - 06/10/2023: Corn, silage

2310431-03

Sample and source description: 2310431-03

Sample date: 10/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,400.00	2,000.00	7,610.00		5.76
DL	0.50	0.50	0.50		0.10

Field 7 - 10/30/2022: Oats, silage-soft dough

2305230-04

Sample and source description: 2305230-04

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	34,700.00	4,390.00	38,900.00		26.30
DL	1.00	0.50	0.50		0.10

2305230-05

Sample and source description: 2305230-05

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 80.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,000.00	2,700.00	24,200.00		21.80
DL	0.50	0.50	0.50		0.10

Field 7 - 06/13/2023: Corn, silage

Field 7 - 06/13/2023: Corn, silage

2310431-04

Sample and source description: 2310431-04

Sample date: 10/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,300.00	1,850.00	8,970.00		5.78
DL	0.50	0.50	0.50		0.10

2310431-05

Sample and source description: 2310431-05

Sample date: 10/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,900.00	1,980.00	10,600.00		6.55
DL	0.50	0.50	0.50		0.10

Field 8 - 10/31/2022: Oats, silage-soft dough

2305230-06

Sample and source description: 2305230-06

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,200.00	2,290.00	19,400.00		18.50
DL	0.50	0.50	0.50		0.10

Field 8 - 06/15/2023: Corn, silage

Field 8 - 06/15/2023: Corn, silage

2310431-06

Sample and source description: 2310431-06

Sample date: 10/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,300.00	1,500.00	5,050.00		4.67
DL	0.50	0.50	0.50		0.10

2310431-07

Sample and source description: 2310431-07

Sample date: 10/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,200.00	2,050.00	9,850.00		5.78
DL	0.50	0.50	0.50		0.10

Field 9 - 11/01/2022: Oats, silage-soft dough

2305231-01

Sample and source description: 2305231-01

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	33,100.00	1,930.00	17,600.00		16.50
DL	0.50	0.50	0.50		0.10

Field 9 - 11/01/2022: Oats, silage-soft dough

2305231-02

Sample and source description: 2305231-02

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,300.00	2,570.00	24,800.00		15.60
DL	0.50	0.50	0.50		0.10

Field 9 - 06/14/2023: Corn, silage

2310431-08

Sample and source description: 2310431-08

Sample date: 10/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,300.00	2,150.00	7,970.00		5.61
DL	0.50	0.50	0.50		0.10

2310431-09

Sample and source description: 2310431-09

Sample date: 10/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,600.00	2,080.00	7,860.00		5.53
DL	0.50	0.50	0.50		0.10

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

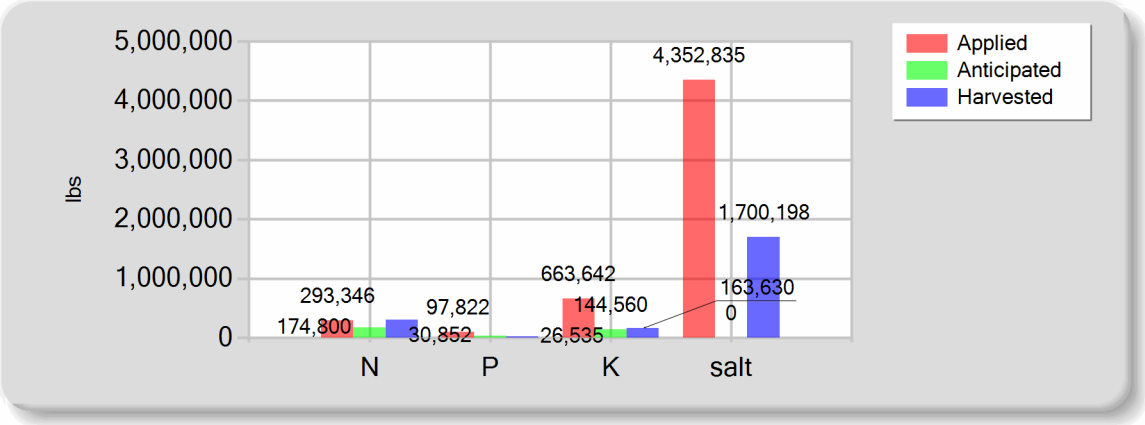
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

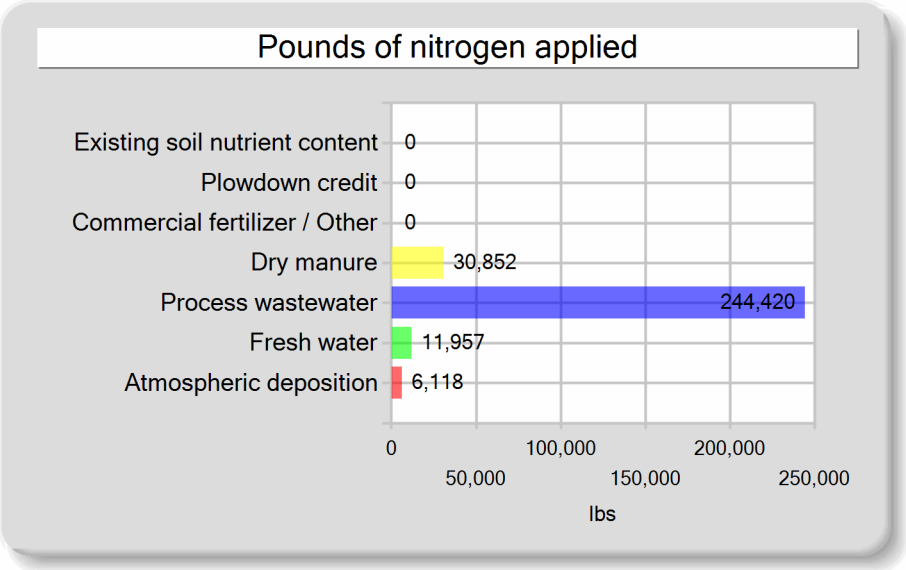
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

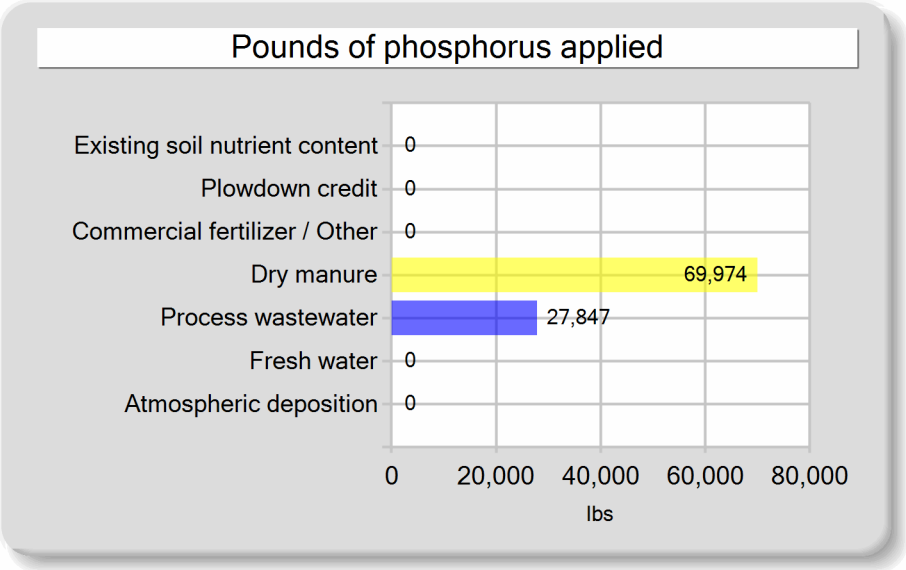
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	30,851.55	69,974.30	386,264.76	2,522,714.17
Process wastewater	244,419.71	27,847.48	277,376.78	1,502,450.49
Fresh water	11,956.95	0.00	0.00	327,669.99
Atmospheric deposition	6,118.00	0.00	0.00	0.00
Total nutrients applied	293,346.21	97,821.78	663,641.54	4,352,834.65
Anticipated crop nutrient removal	174,800.00	30,852.20	144,559.60	0.00
Actual crop nutrient removal	300,592.17	26,535.26	163,629.92	1,700,198.33
Nutrient balance	-7,245.96	71,286.52	500,011.62	2,652,636.31
Applied to removed ratio	0.98	3.69	4.06	2.56

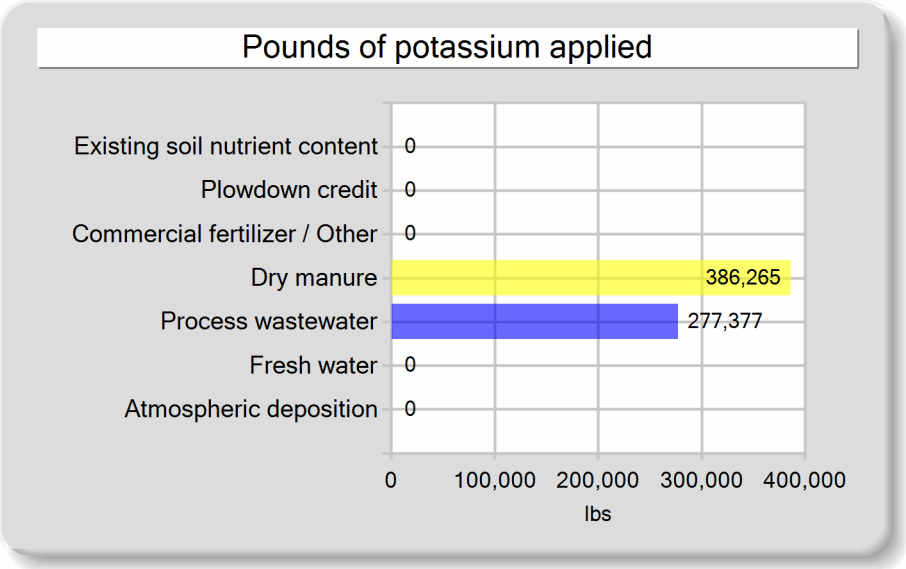
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

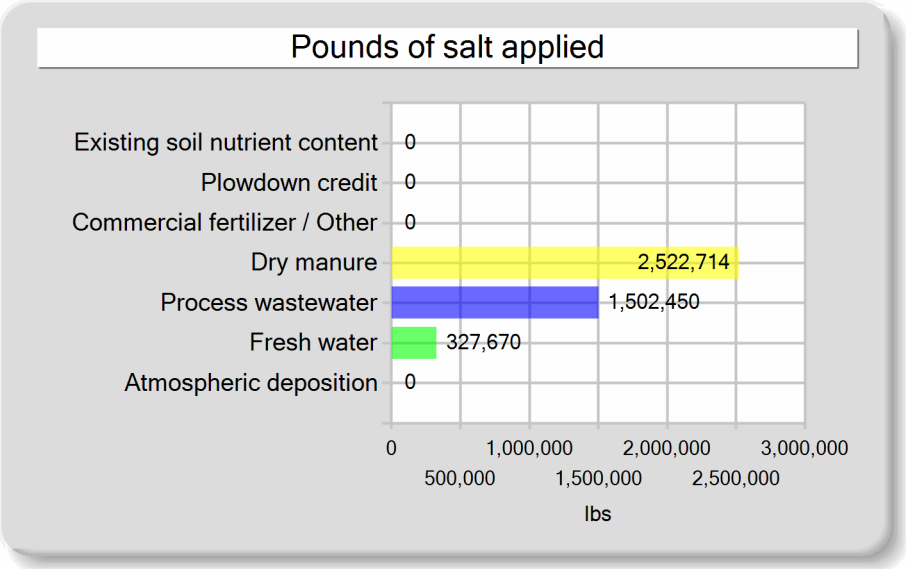


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Farms LLC Da Silva	Dennis DaSilva
PRINT OR TYPE NAME	PRINT OR TYPE NAME
07/02/2024	07/02/2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment	N/A
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .	

Manure/Process Wastewater Tracking Manifests	N/A
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.	

Corrective Actions Documents	N/A
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.	

Groundwater Monitoring	N/A
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.	

Storm Water Monitoring	N/A
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.	